

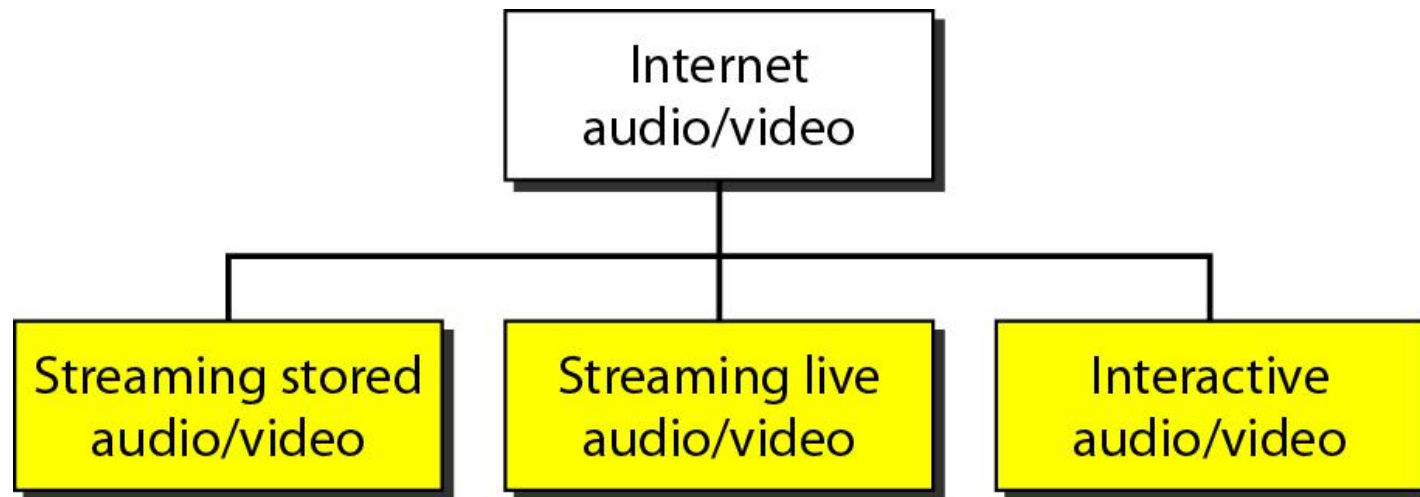
Lecture 11

Multimedia

Outline

- **11-1 DIGITIZING AUDIO AND VIDEO**
- **11-2 AUDIO AND VIDEO COMPRESSION**
- **11-3 STREAMING STORED AUDIO/VIDEO**
- **11-4 STREAMING LIVE AUDIO/VIDEO**
- **11-5 REAL-TIME INTERACTIVE AUDIO/VIDEO**
- **11-6 RTP**
- **11-7 RTCP**
- **11-8 VOICE OVER IP**

Figure 11.1 *Internet audio/video*





Note

Streaming stored audio/video refers to on-demand requests for compressed audio/video files.



Note

Streaming live audio/video refers to the broadcasting of radio and TV programs through the Internet.



Note

Interactive audio/video refers to the use of the Internet for interactive audio/video applications.

11-1 DIGITIZING AUDIO AND VIDEO

Before audio or video signals can be sent on the Internet, they need to be digitized. We discuss audio and video separately.

Topics discussed in this section:

Digitizing Audio

Digitizing Video



Note

**Compression is needed to send video
over the Internet.**

11-2 AUDIO AND VIDEO COMPRESSION

To send audio or video over the Internet requires compression. In this section, we discuss audio compression first and then video compression.

JPEG: Joint Photographic Expert Groups

MPRG: Moving Picture Experts Group

Topics discussed in this section:

Audio Compression

Video Compression

Figure 11.2 *JPEG gray scale*

Gray scale: each pixel can be represented by an 8-bit integer (256 levels). In JPEG, a gray scale picture is divided into blocks of 8×8 pixels. The purpose of dividing the picture into blocks is to decrease the number of calculations.

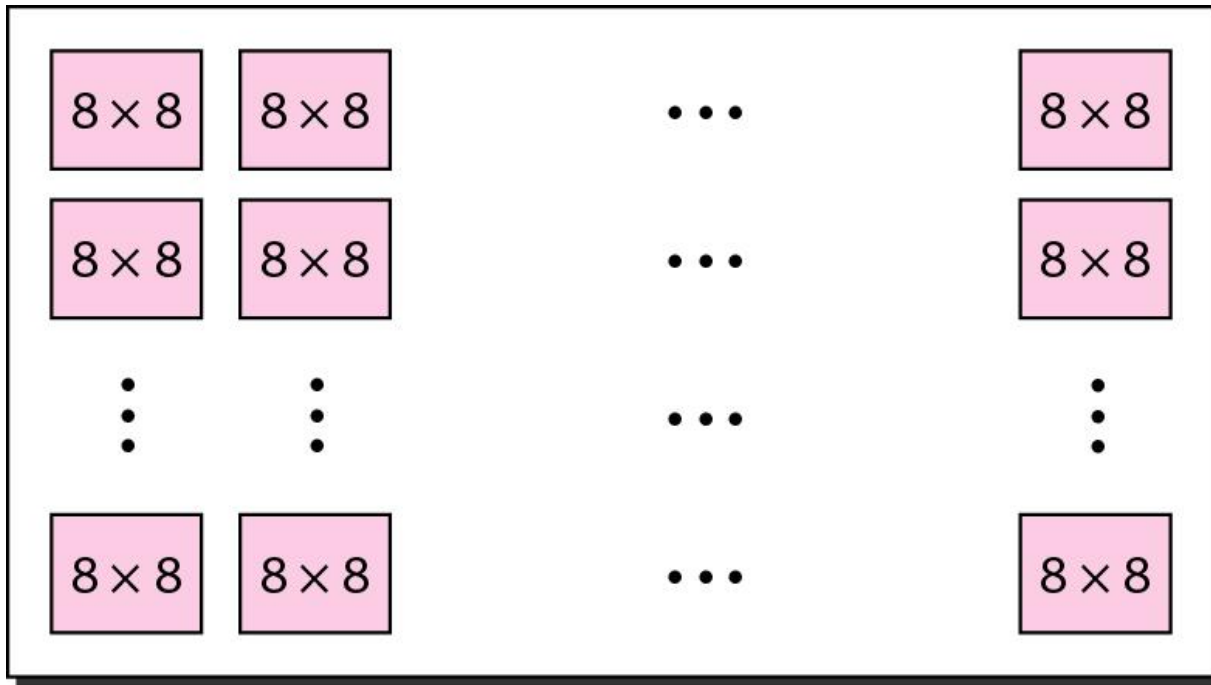


Figure 11.3 *JPEG process*

Discrete Cosine Transform (DCT): each block of 64pixels goes through a transformation called the DCT.

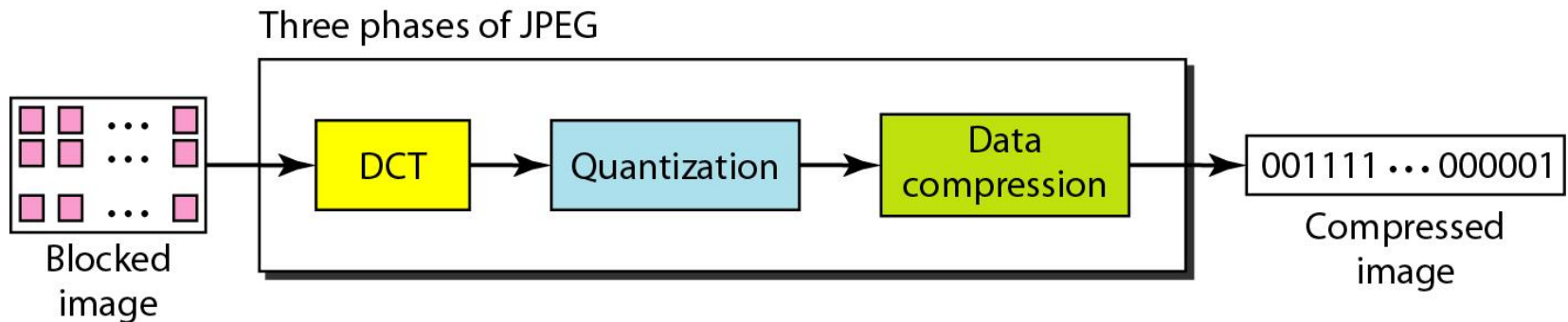


Figure 11.4 *Case 1: uniform gray scale*

The value $T(0,0)$ is the average (multiplied by a constant) of the $P(x,y)$ values and is called the dc value (direct current). The rest of the values, called ac values, in $T(m,n)$ represent changes in the pixel.

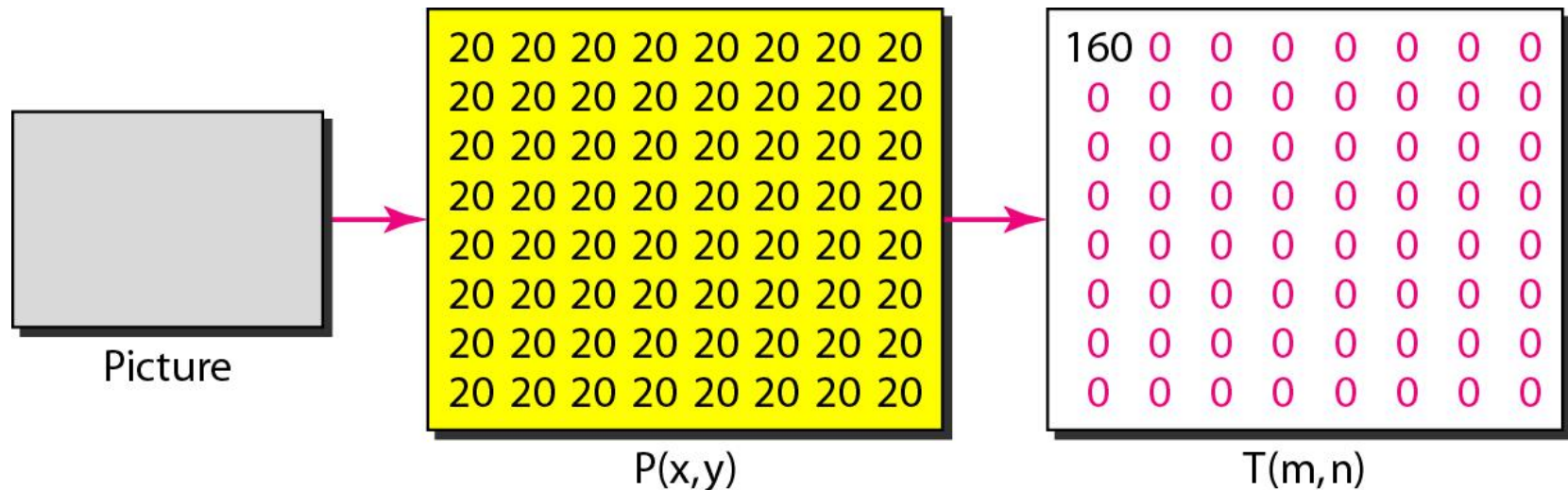


Figure 11.5 *Case 2: two sections*

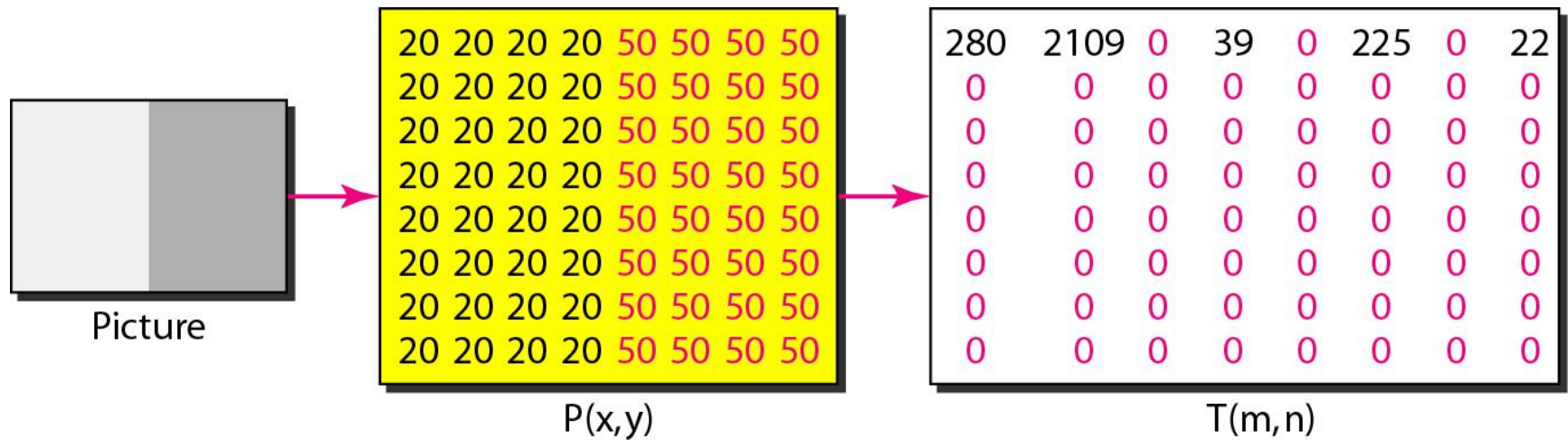


Figure 11.6 *Case 3: gradient gray scale*

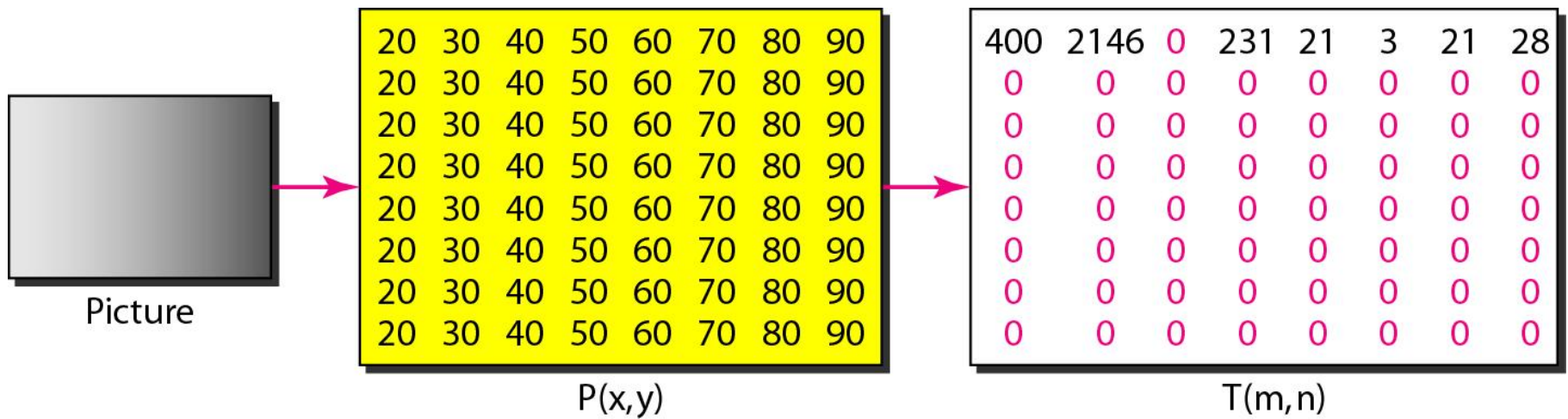


Figure 11.7 *Reading the table*

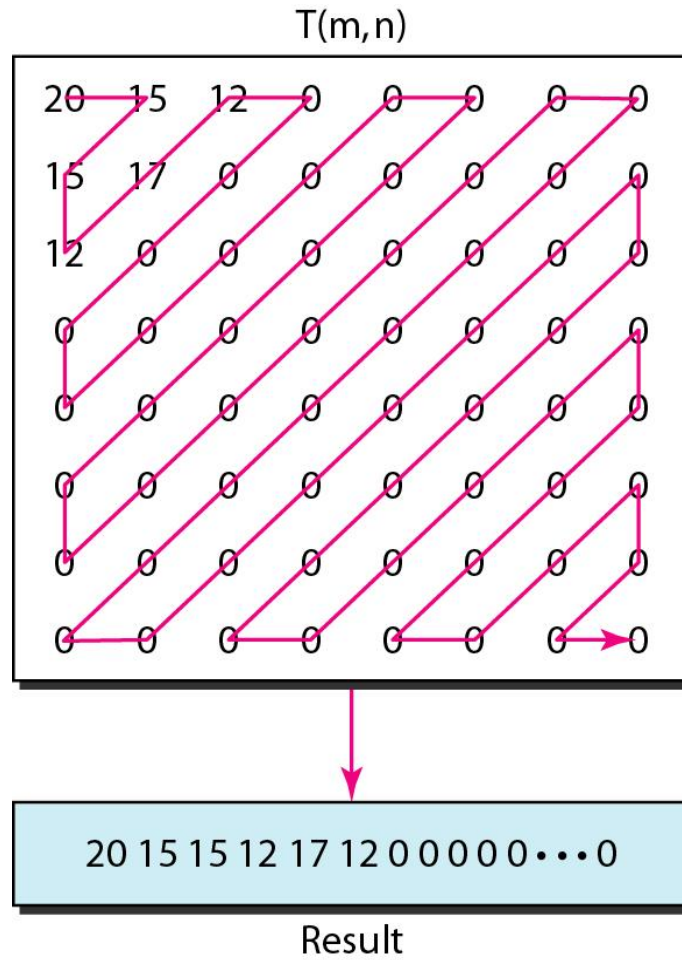


Figure 11.8 *MPEG frames*

Temporal Compression: redundant frames are removed.

I-frames: An intracoded frame is an independent frame that is not related to any other frame.

P-frames: A predicted frame is related to the preceding I-frame or P-frame.

B-frames: A bidirectional frame is related to the preceding and following I-frame or P-frame. Each B-frame is relative to the past and the future.

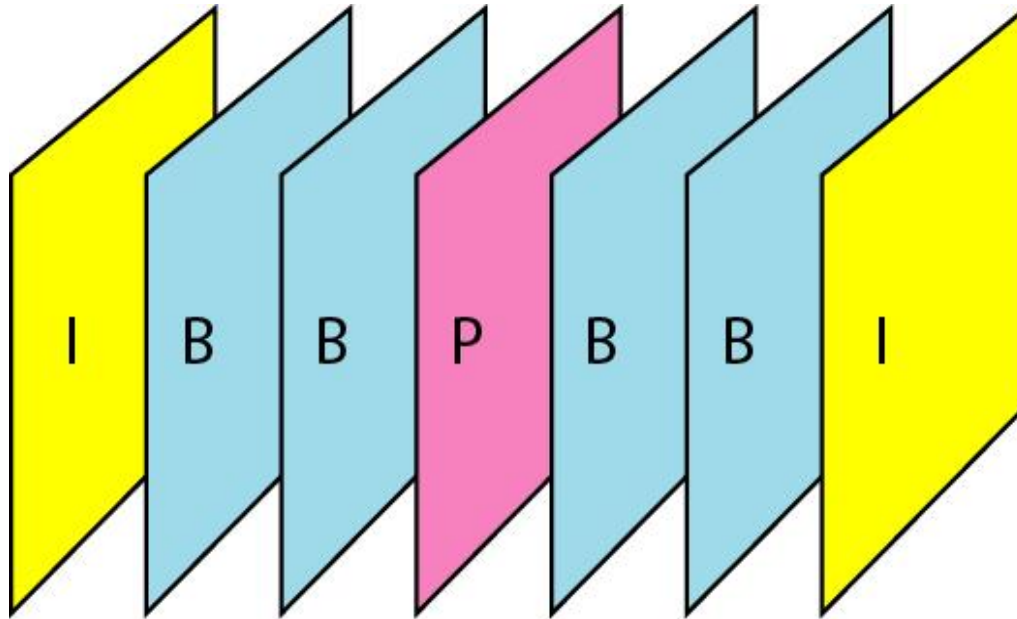
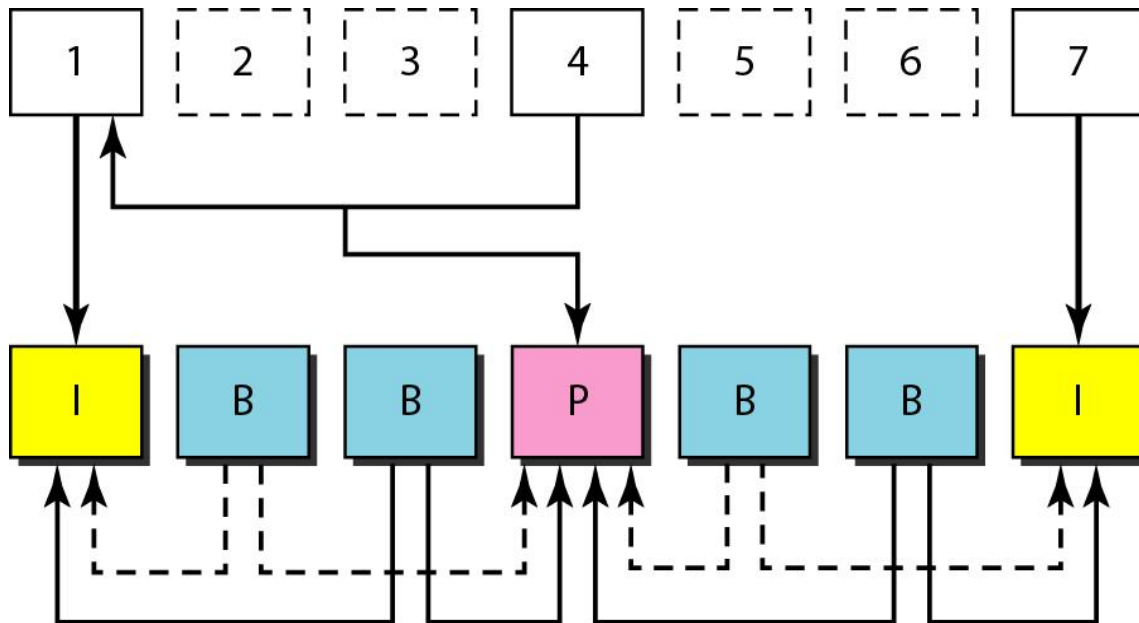


Figure 11.9 *MPEG frame construction*



11-3 STREAMING STORED AUDIO/VIDEO

Now that we have discussed digitizing and compressing audio/video, we turn our attention to specific applications. The first is streaming stored audio and video.

Topics discussed in this section:

First Approach: Using a Web Server

Second Approach: Using a Web Server with a Metafile

Third Approach: Using a Media Server

Fourth Approach: Using a Media Server and RTSP

Figure 11.10 *Using a Web server*

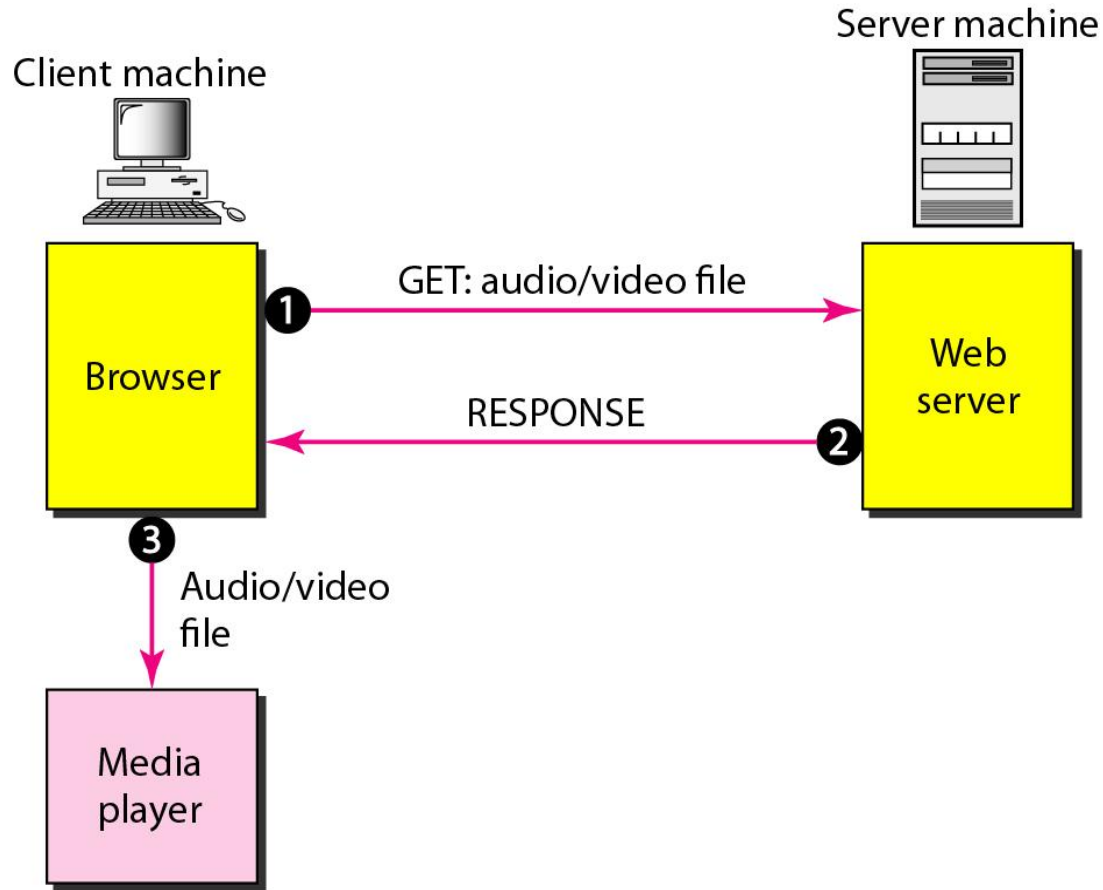


Figure 11.11 *Using a Web server with a metafile*

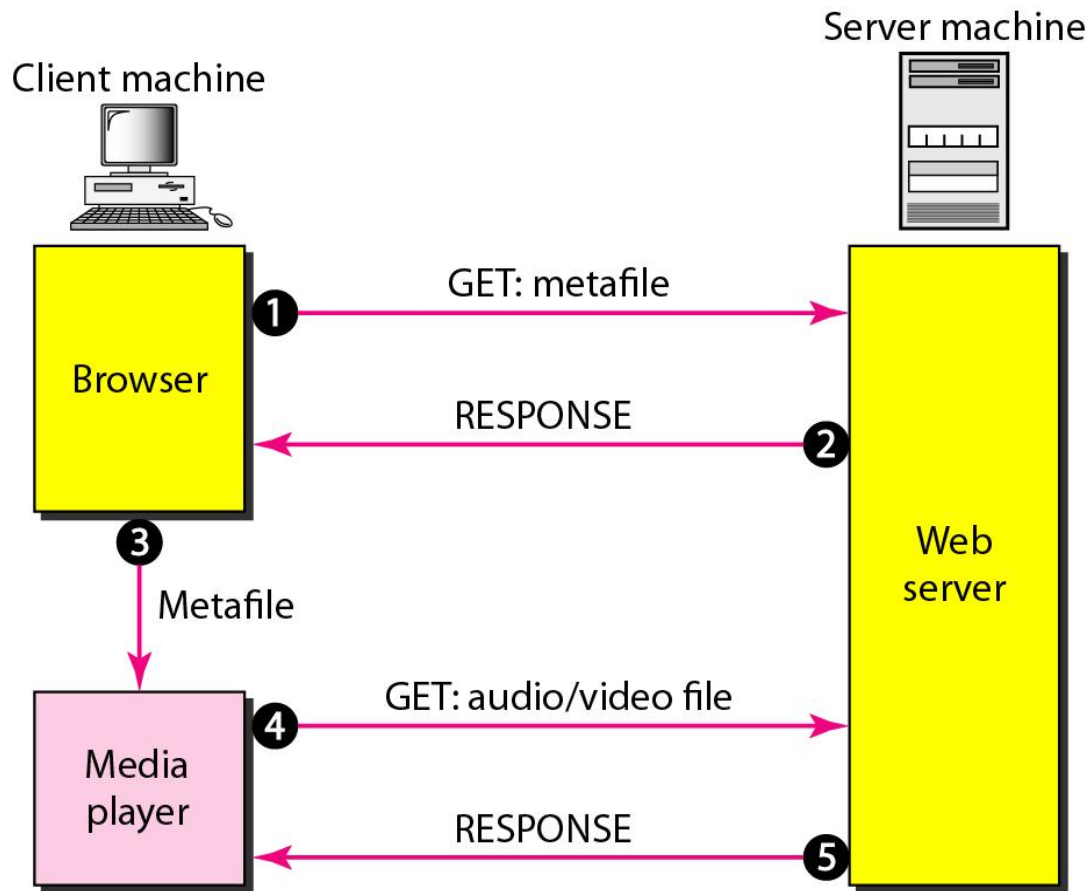


Figure 11.12 *Using a media server*

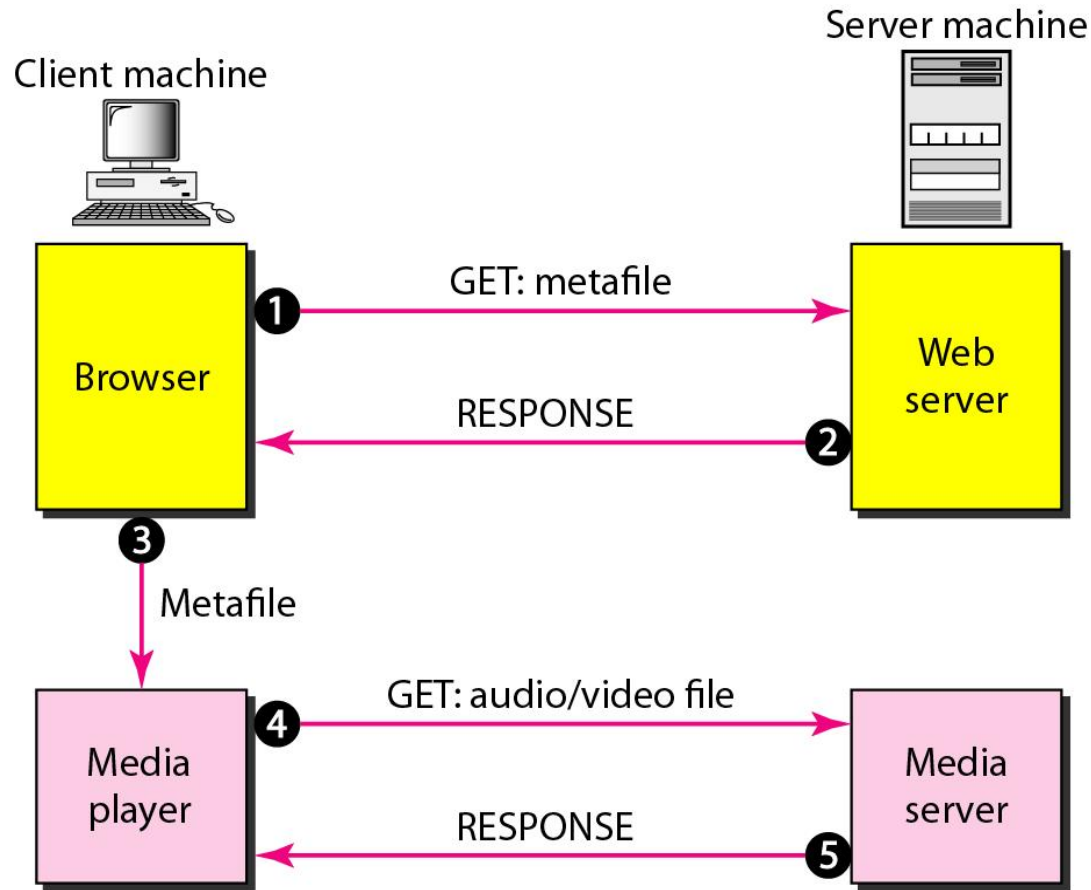
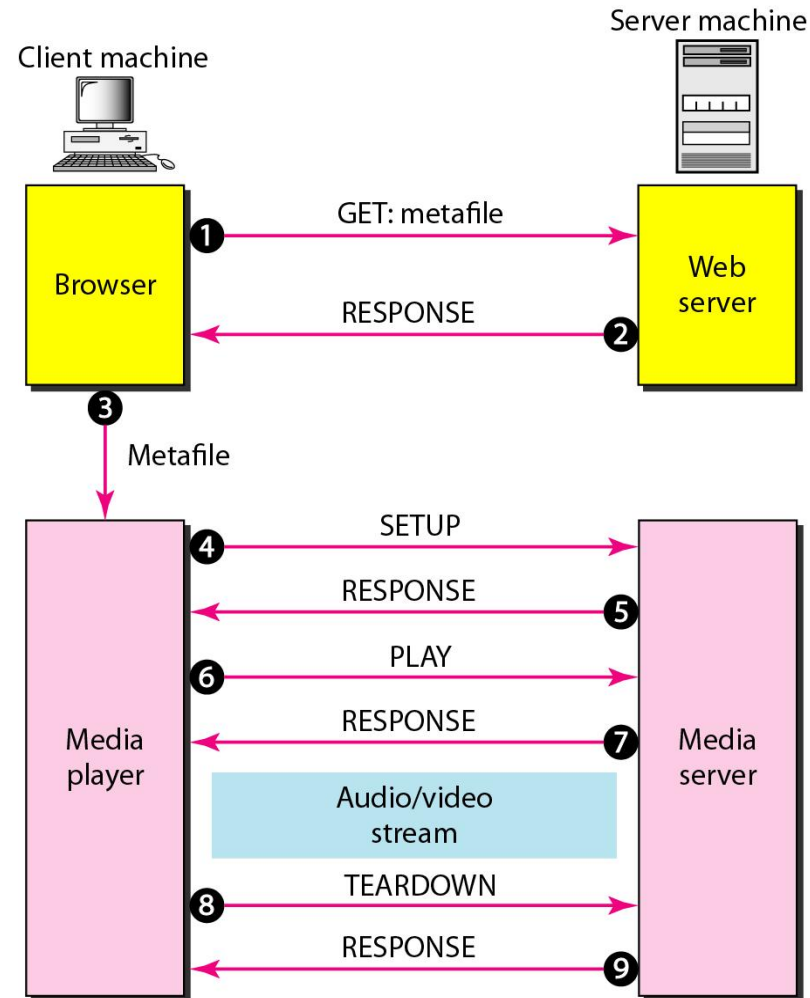


Figure 11.13 *Using a media server and RTSP*



11-4 STREAMING LIVE AUDIO/VIDEO

Streaming live audio/video is similar to the broadcasting of audio and video by radio and TV stations. Instead of broadcasting to the air, the stations broadcast through the Internet. There are several similarities between streaming stored audio/video and streaming live audio/video. They are both sensitive to delay; neither can accept retransmission. However, there is a difference. In the first application, the communication is unicast and on-demand. In the second, the communication is multicast and live.

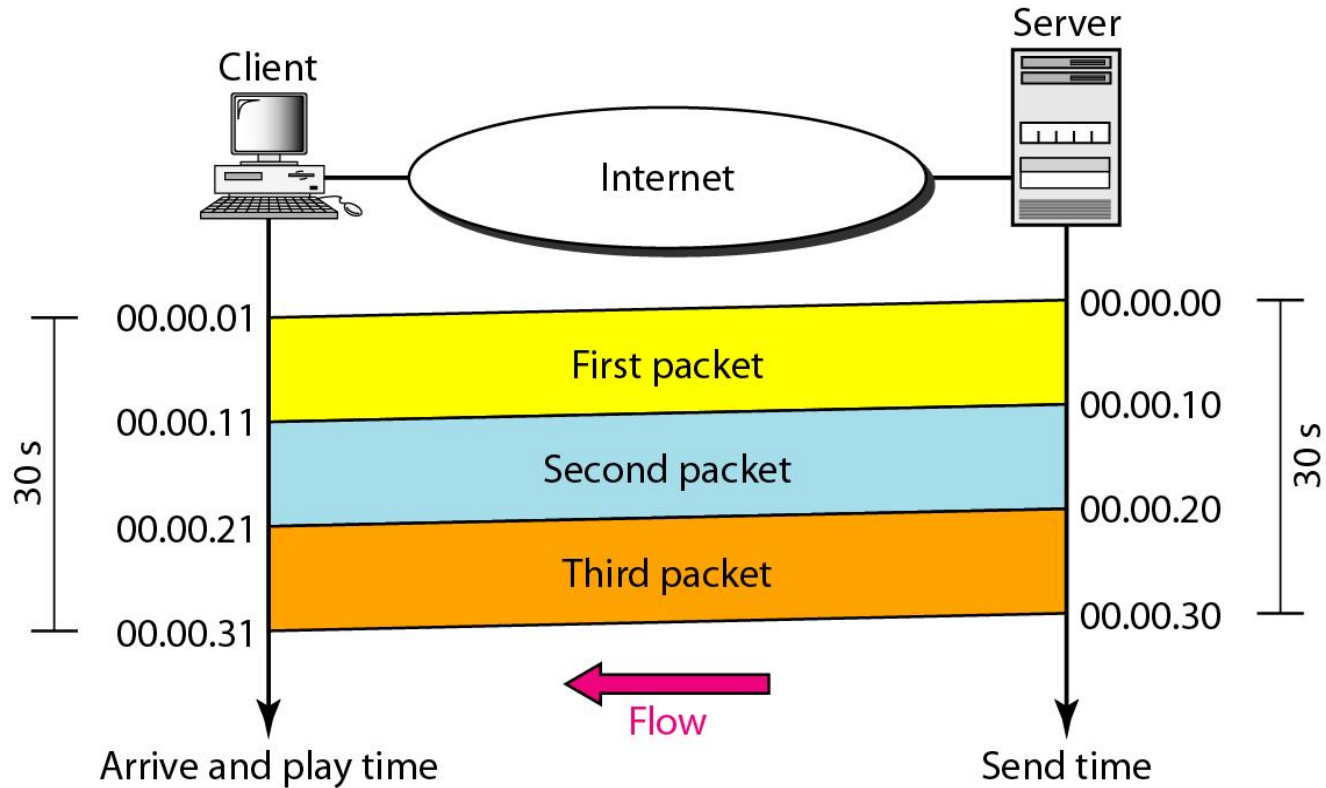
11-5 REAL-TIME INTERACTIVE AUDIO/VIDEO

In real-time interactive audio/video, people communicate with one another in real time. The Internet phone or voice over IP is an example of this type of application. Video conferencing is another example that allows people to communicate visually and orally.

Topics discussed in this section:

Characteristics

Figure 11.14 *Time relationship*





Note

Jitter is introduced in real-time data by the delay between packets.

Figure 11.15 *Jitter*

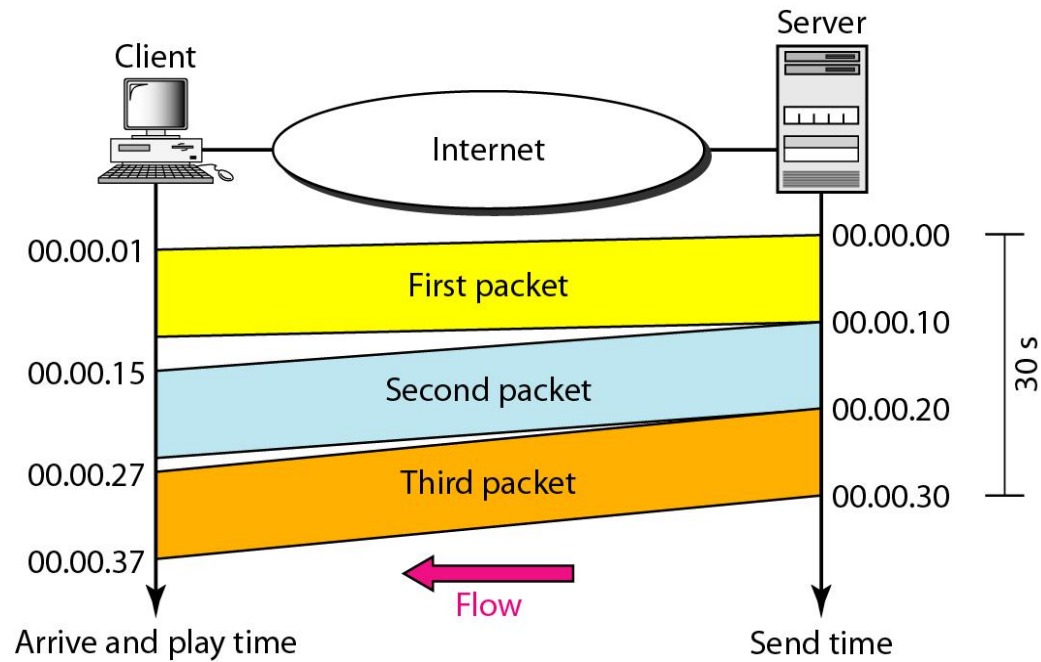
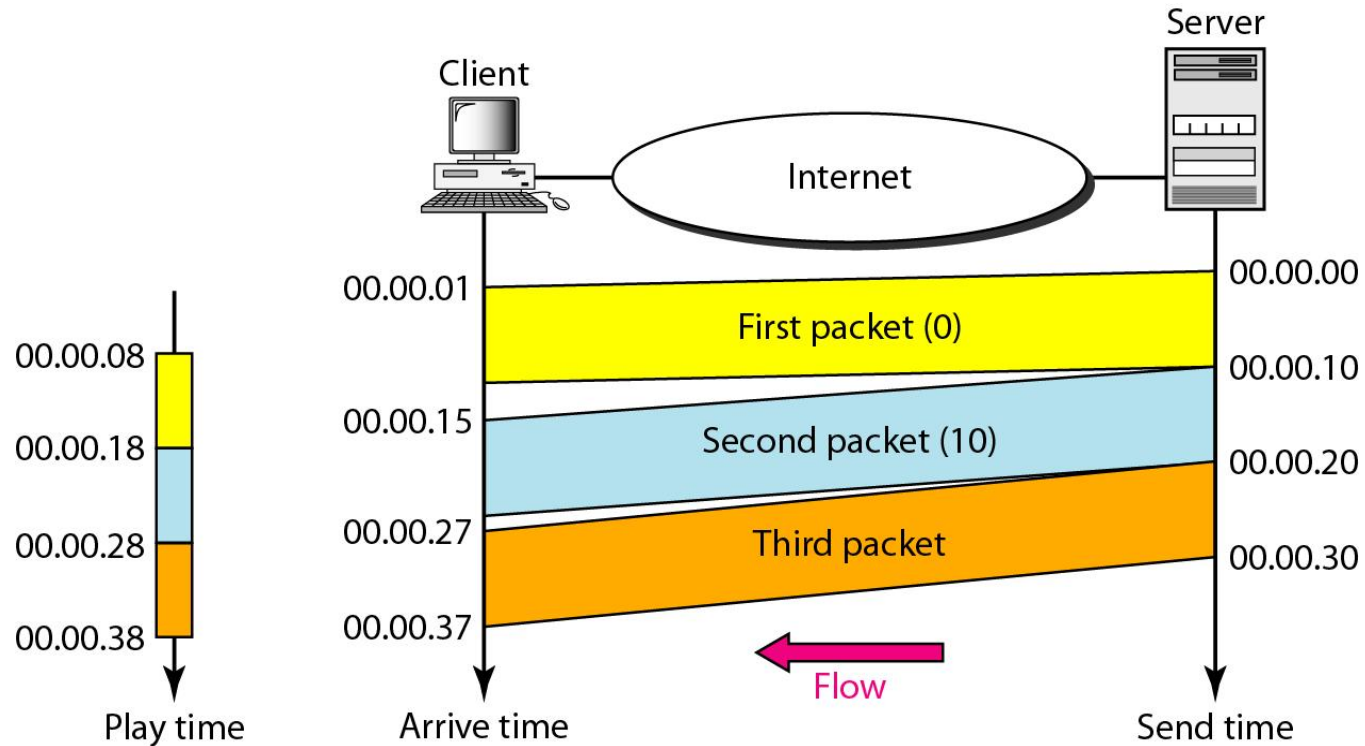


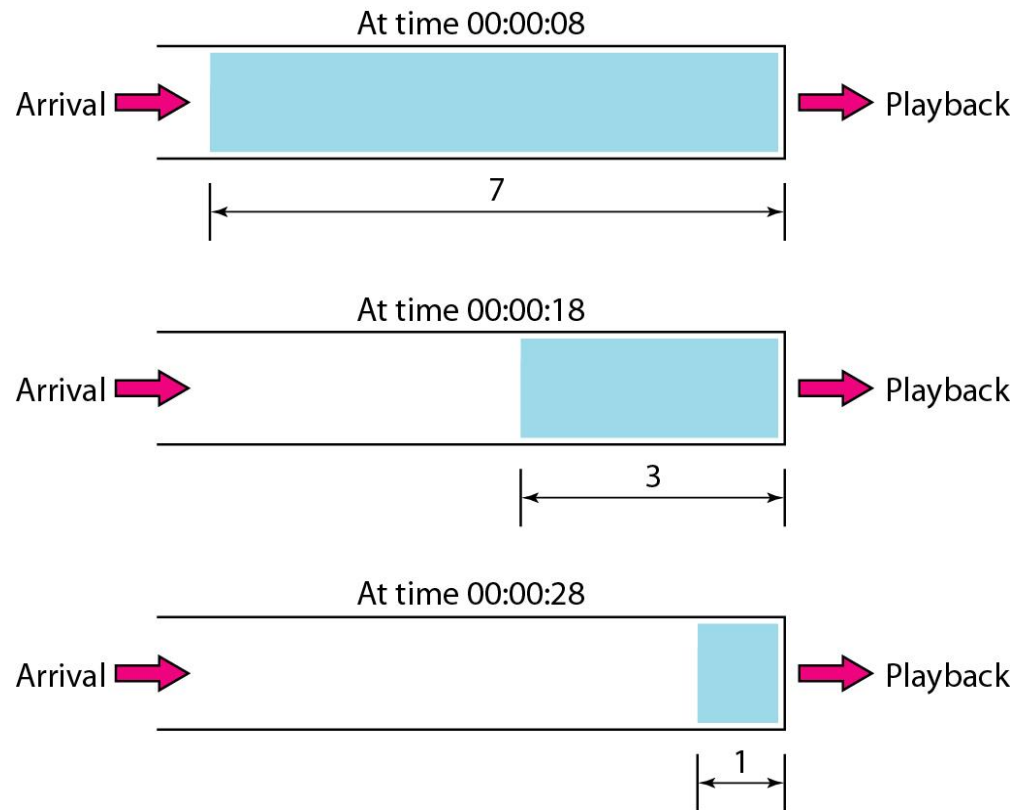
Figure 11.16 *Timestamp*



Note

To prevent jitter, we can time-stamp the packets and separate the arrival time from the playback time.

Figure 11.17 *Playback buffer*





Note

A playback buffer is required for real-time traffic.



Note

A sequence number on each packet is required for real-time traffic.



Note

Real-time traffic needs the support of multicasting.

Note

Translation means changing the encoding of a payload to a lower quality to match the bandwidth of the receiving network.



Note

Mixing means combining several streams of traffic into one stream.



Note

TCP, with all its sophistication, is not suitable for interactive multimedia traffic because we cannot allow retransmission of packets.



Note

UDP is more suitable than TCP for interactive traffic. However, we need the services of RTP, another transport layer protocol, to make up for the deficiencies of UDP.

11-6 RTP

Real-time Transport Protocol (RTP) is the protocol designed to handle real-time traffic on the Internet. RTP does not have a delivery mechanism; it must be used with UDP. RTP stands between UDP and the application program. The main contributions of RTP are time-stamping, sequencing, and mixing facilities.

Topics discussed in this section:

RTP Packet Format

UDP Port

Figure 11.18 *RTP*

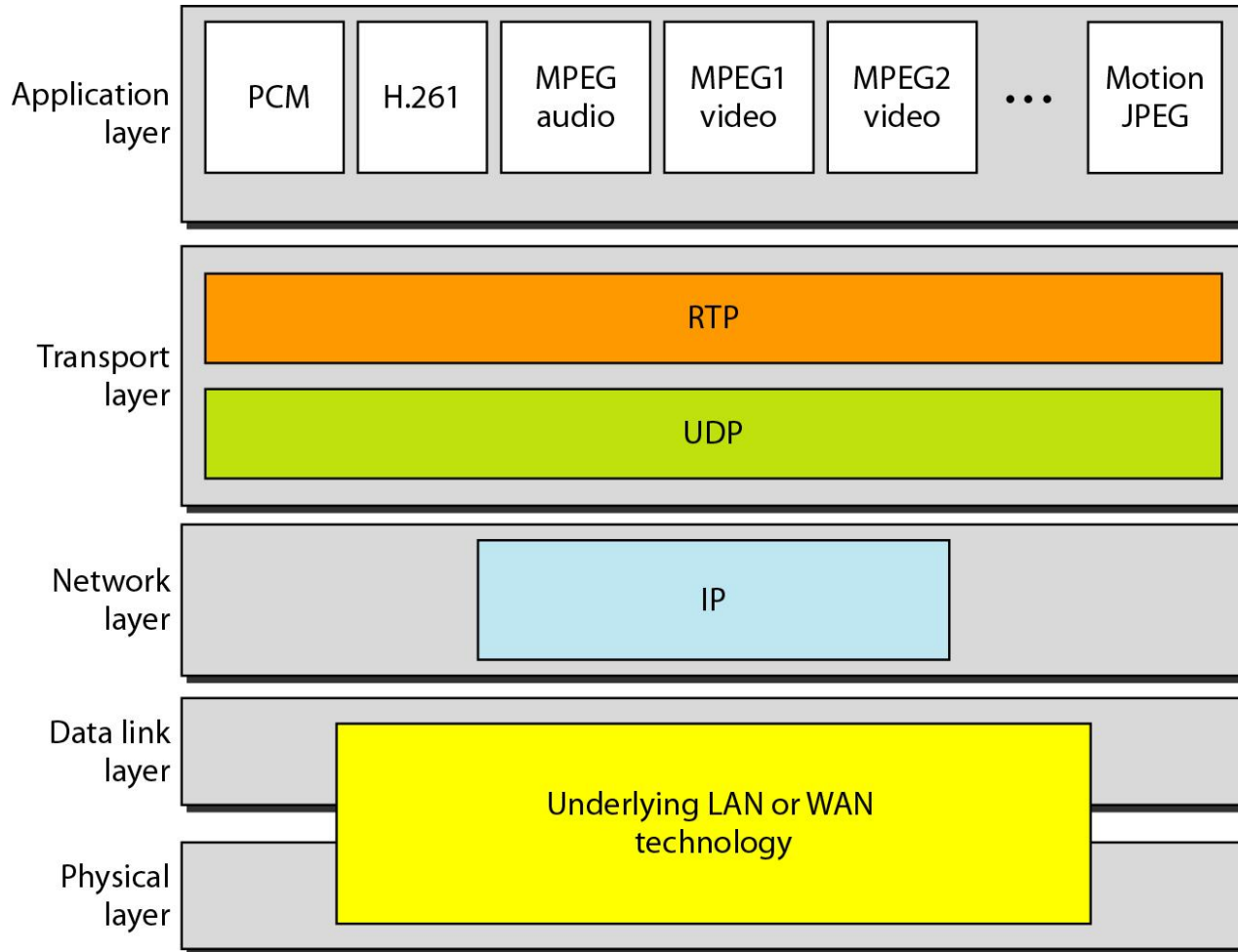


Figure 11.19 *RTP packet header format*

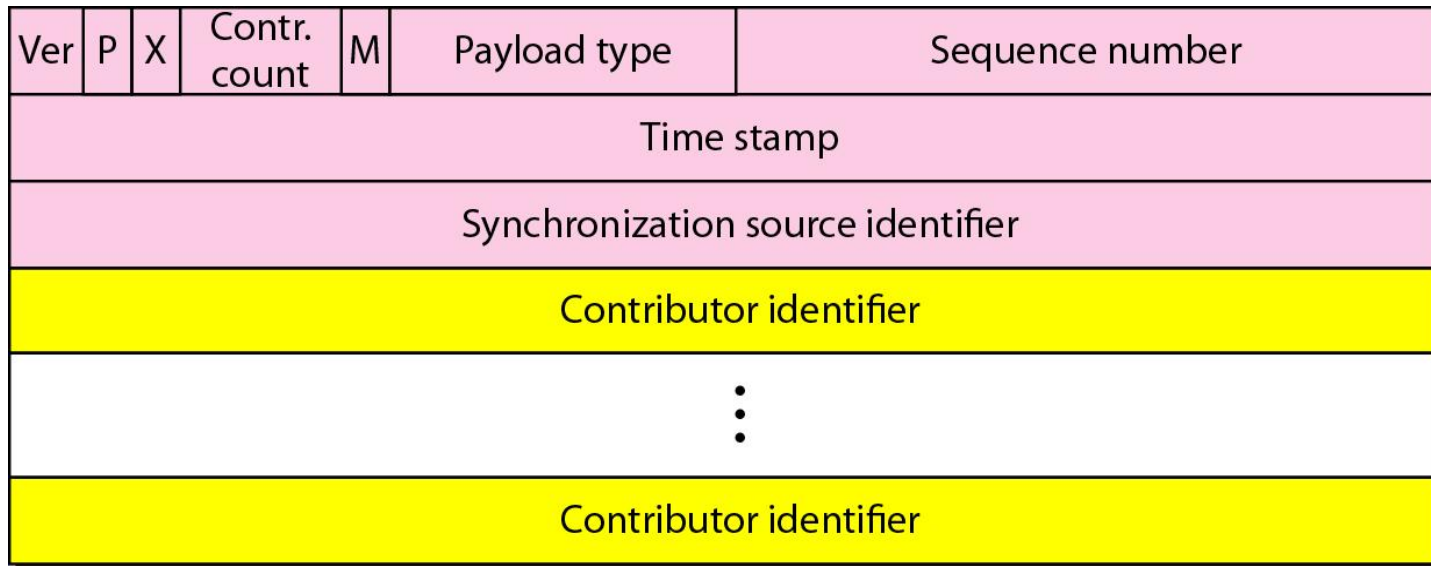


Table 20.1 *Payload types*

<i>Type</i>	<i>Application</i>	<i>Type</i>	<i>Application</i>	<i>Type</i>	<i>Application</i>
0	PCMu Audio	7	LPC audio	15	G728 audio
1	1016	8	PCMA audio	26	Motion JPEG
2	G721 audio	9	G722 audio	31	H.261
3	GSM audio	10–11	L16 audio	32	MPEG1 video
5–6	DV14 audio	14	MPEG audio	33	MPEG2 video



Note

**RTP uses a temporary even-numbered
UDP port.**

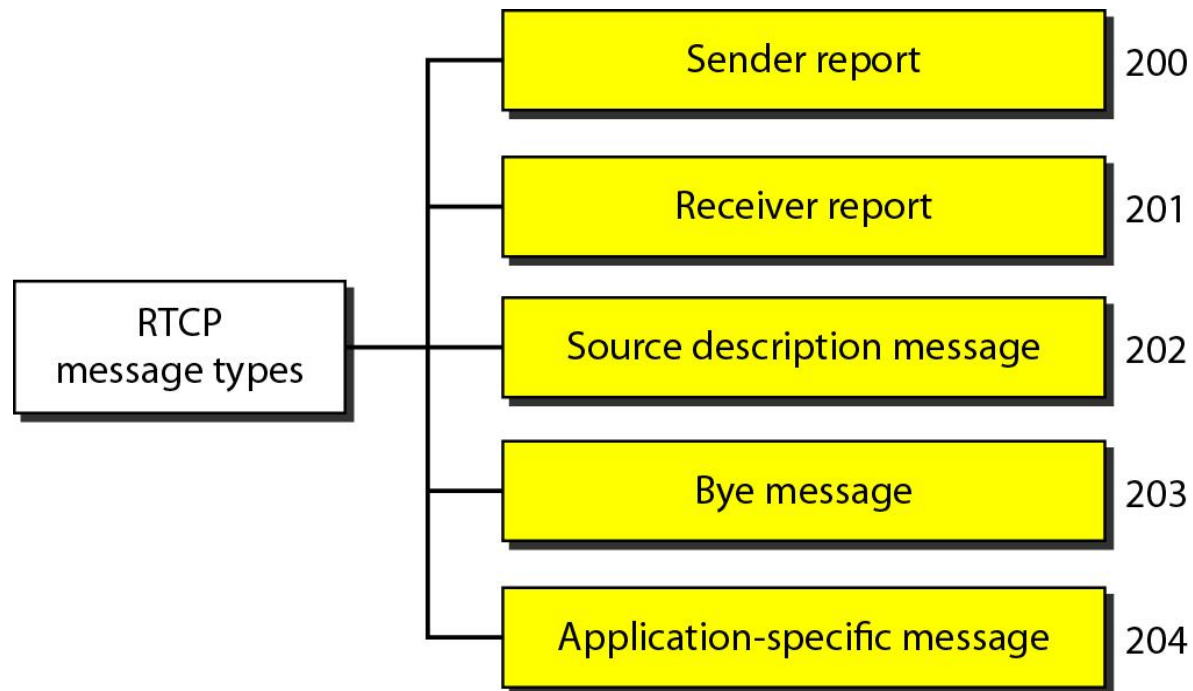
11-7 RTCP

RTP allows only one type of message, one that carries data from the source to the destination. In many cases, there is a need for other messages in a session. These messages control the flow and quality of data and allow the recipient to send feedback to the source or sources. Real-time Transport Control Protocol (RTCP) is a protocol designed for this purpose.

Topics discussed in this section:

Sender Report and Receiver Report
Messages
UDP Port

Figure 11.20 *RTCP message types*





Note

RTCP uses an odd-numbered UDP port number that follows the port number selected for RTP.

11-8 VOICE OVER IP

Let us concentrate on one real-time interactive audio/video application: voice over IP, or Internet telephony. The idea is to use the Internet as a telephone network with some additional capabilities. Two protocols have been designed to handle this type of communication: Session Initiation Protocol (SIP) and H.323.

Topics discussed in this section:

SIP

H.323

Figure 11.21 *SIP messages*

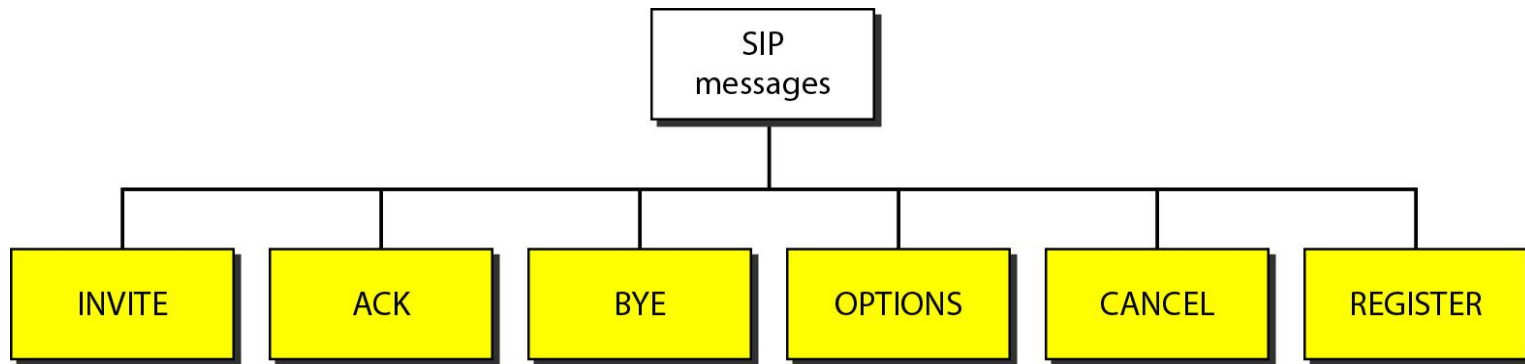


Figure 11.22 *SIP formats*

SIP is very flexible. In SIP, an email address, an IP address, a telephone number, and other types of addresses can be used to identify the sender and receiver. However, the address needs to be in SIP format.

sip:bob@201.23.45.78

IPv4 address

sip:bob@fhda.edu

E-mail address

sip:bob@408-864-8900

Phone number

Figure 11.23 *SIP simple session*

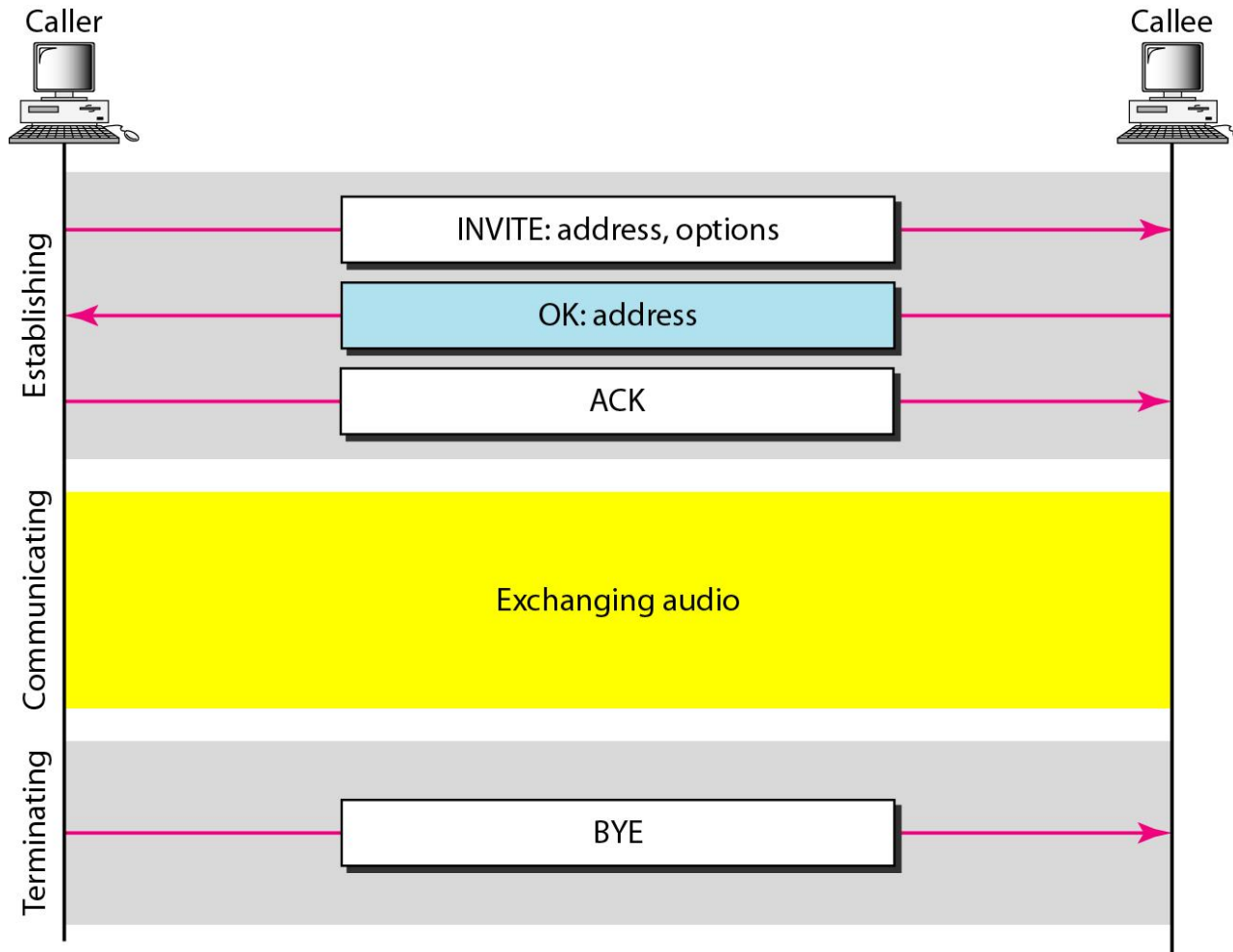


Figure 11.24 *Tracking the callee*

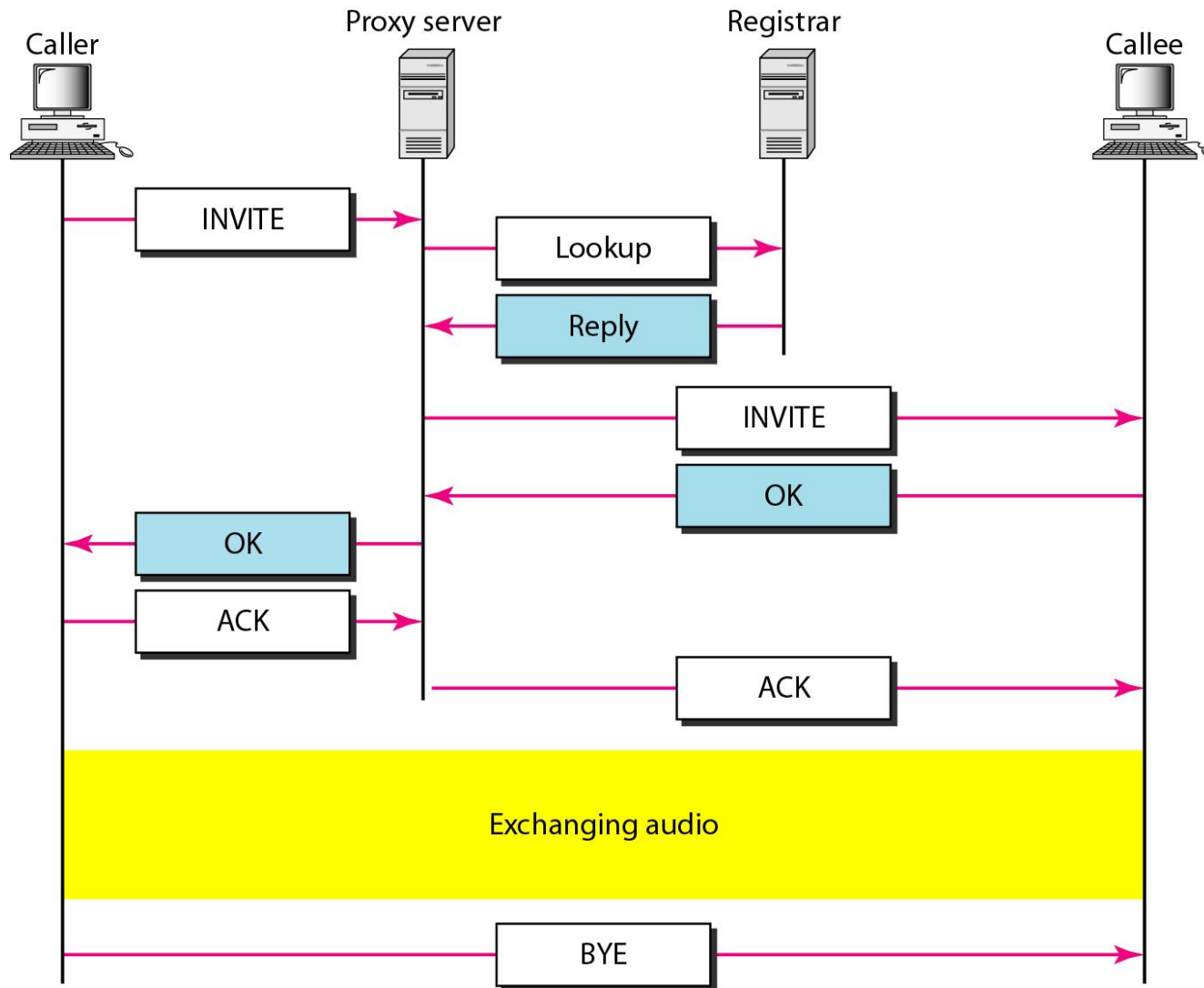


Figure 11.25 *H.323 architecture*

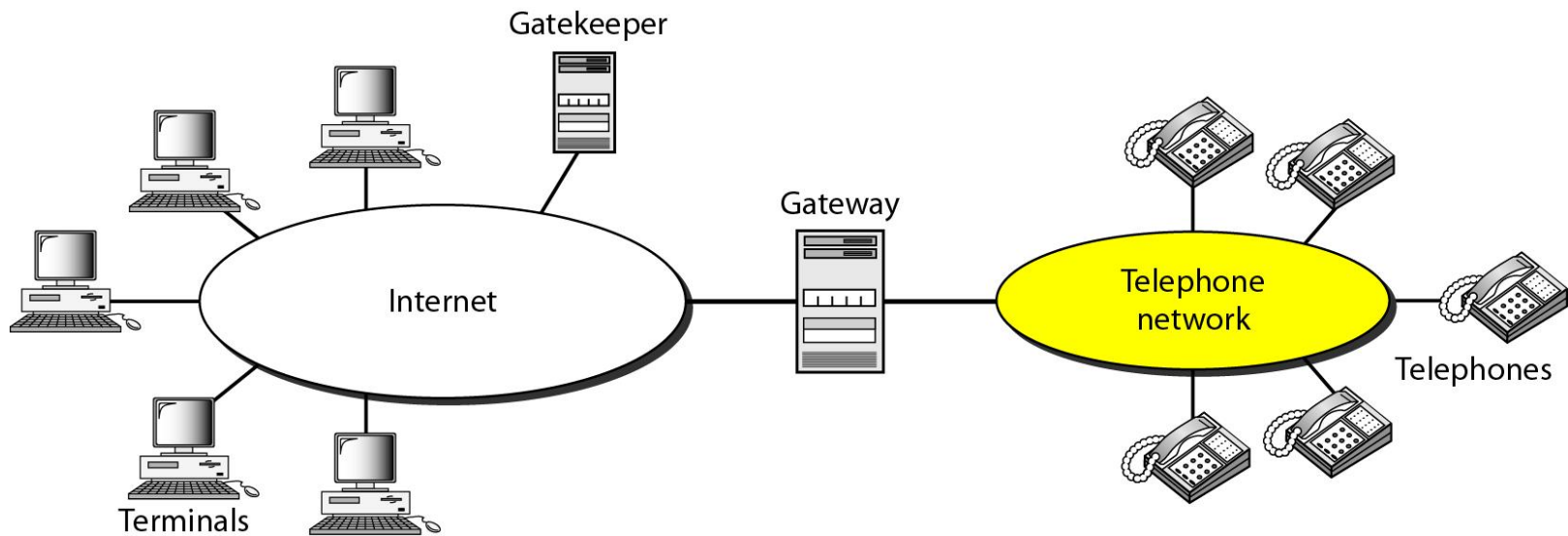


Figure 11.26 *H.323 protocols*

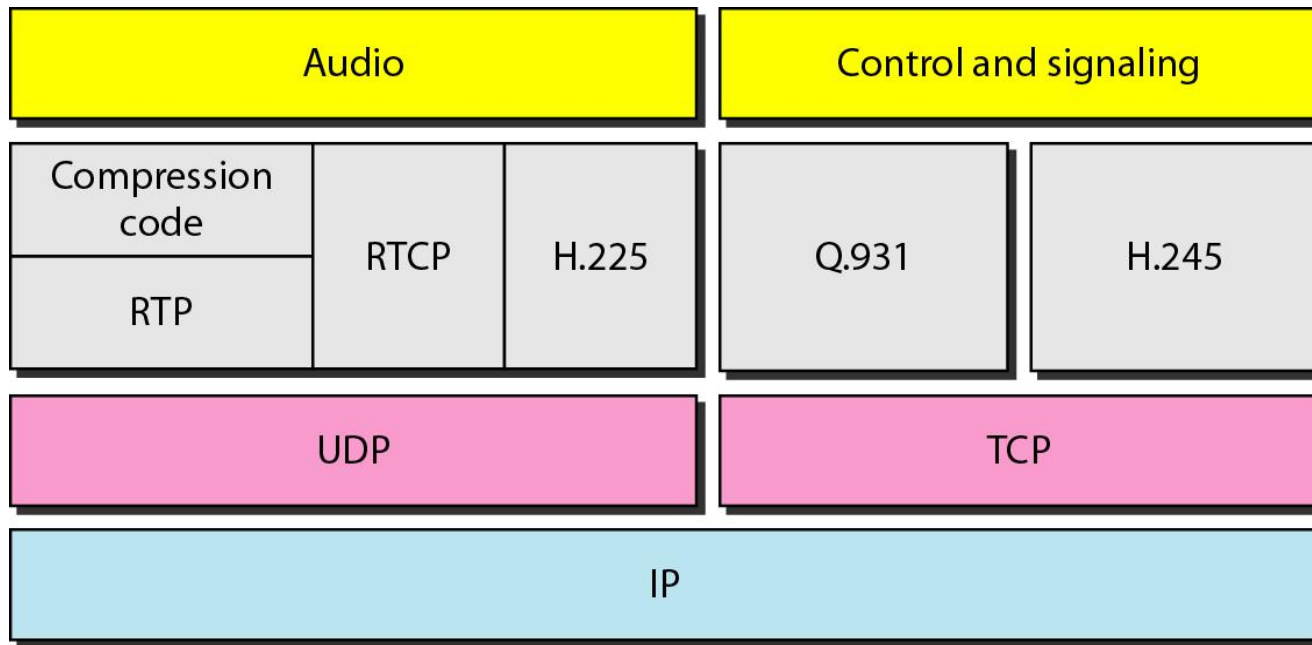
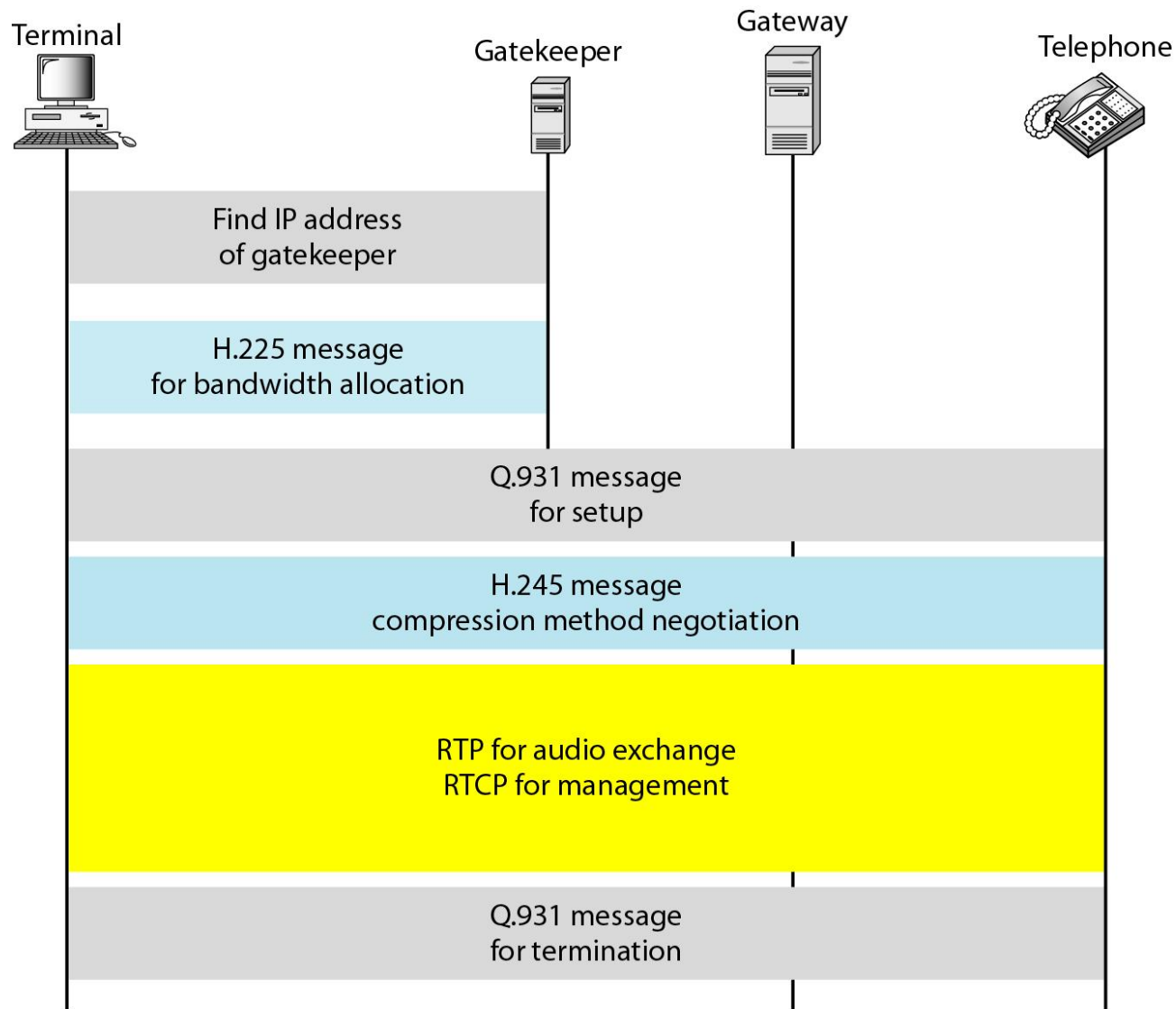


Figure 11.27 *H.323 example*



Summary

- Audio/video files can be downloaded for future use (streaming stored audio/video) or broadcast to clients over the Internet (streaming live audio/video). The Internet can also be used for live audio/video interaction.
- Audio and video need to be digitized before being sent over the Internet.
- Audio files are compressed through predictive encoding or perceptual encoding.
- Joint Photographic Experts Group (JPEG) is a method to compress pictures and graphics.
- The JPEG process involves blocking, the discrete cosine transform, quantization, and lossless compression.

Summary-Cont.

- Moving Pictures Experts Group (MPEG) is a method to compress video.
- MPEG involves both spatial compression and temporal compression. The former is similar to JPEG, and the latter removes redundant frames.
- We can use a Web server, or a Web server with a metafile, or a media server, or a media server and RSTP to download a streaming audio/video file.
- Real-time data on a packet-switched network require the preservation of the time relationship between packets of a session.
- Gaps between consecutive packets at the receiver cause a phenomenon called jitter.

Summary-Cont.

- Jitter can be controlled through the use of timestamps and a judicious choice of the playback time.
- A playback buffer holds data until they can be played back.
- A receiver delays playing back real-time data held in the playback buffer until a threshold level is reached.
- Sequence numbers on real-time data packets provide a form of error control.
- Real-time data are multicast to receivers.

Summary-Cont.

- Real-time data traffic sometimes requires a translator to change a high-bandwidth signal to a lower-quality narrow-bandwidth signal.
- A mixer combines signals from different sources into one signal.
- Real-time multimedia traffic requires both UDP and Real-Time Transport Protocol (RTP).
- RTP handles timestamping, sequencing, and mixing.
- Real-time Transport Control Protocol (RTCP) provides flow control, quality of data control, and feedback to the sources.

Summary-Cont.

- Voice over IP is a real-time interactive audio/video application.
- The Session Initiation Protocol (SIP) is an application layer protocol that establishes, manages, and terminates multimedia sessions.
- H.323 is an ITU standard that allows a telephone connected to a public telephone network to talk to a computer connected to the Internet.